

Self-Tuned Rejection Sampling within Gibbs and a Case Study in Small Area Estimation

Guide to Replication Materials

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1 Introduction

This document describes replication materials for the article “Self-Tuned Rejection Sampling within Gibbs and a Case Study in Small Area Estimation”. It includes an R package, code for two simulations, and a data analysis. The following folders are included in the materials.

- **01-sim-cond**: simulation generating from a single conditional of the SAE model under consideration.
- **02-sim-post**: simulation generating from the posterior of the SAE model under consideration.
- **03-sae**: data analysis with the SAIPE dataset.
- **data**: the SAIPE dataset and codes /inputs to assemble it.
- **vws**: An R package that supports construction of VWS samplers.
- **saevws**: an R package for the Gibbs sampler and supporting functions. It can be configured with basic vertical weighted strips (VWS), self-tuned VWS, and independent Metropolis-Hastings (IMH) for the $\sigma_1^2, \dots, \sigma_m^2$ steps.

A working installation of R ([R Core Team 2026](#)) is required with the following packages (and their dependencies). These packages are available on CRAN at the time of this writing.

- **devtools** ([Wickham et al. 2022](#)),
- **fnt1** ([Raim 2024](#))
- **jsonlite** ([Ooms 2014](#))
- **knitr** ([Xie 2014](#))
- **mcmcse** ([Flegal et al. 2021](#))
- **Rcpp** ([Eddelbuettel and Balamuta 2018](#)),
- **statmod** ([Smyth 2005](#))
- **tidyverse** ([Wickham et al. 2019](#)),
- **xtable** ([Dahl et al. 2019](#))

Additionally, the included **vws** package must be installed.

```
R> devtools::install_local("/path/to/vws")
```

Section 2 gives an overview of the **saevws** package, including installation and basic usage. Section 3 lists the scripts used for the conditional simulation, posterior simulation, and data analysis. Section 4 gives a comprehensive list of packages and versions used in the computational studies, for reproducibility.

2 saevws R Package

The `saevws` package can be installed using an R command like the following.

```
R> devtools::install_local("/path/to/saevws")
```

TBD: Needs update The package exports several functions for use in R:

- `gibbs_joint` invokes the Gibbs sampler.
- `control_joint` constructs a control object with additional arguments for the Gibbs sampler.
- `get_vws_control` constructs a control object with additional arguments for VWS within the Gibbs sampler.
- `get_init` constructs an object with initial values for the Gibbs sampler.
- `get_fixed` constructs an object to indicate that certain elements of the MCMC chain are to remain fixed in the Gibbs sampler. This is primarily used for testing the code.
- `traceplots` is a utility to make trace plots from a matrix with draws.

These functions' interfaces are shown below; see the associated manual pages for details about arguments, outputs, etc..

```
gibbs_joint(y, s2, X, Z, df, init = init_joint(m = length(y), d1 = ncol(X), d2 =  
  ncol(Z)), control = control_joint(), fixed = fixed_joint())
```

```
control_joint(R = 1000, burn = 0, thin = 1, report = R + 1, save_latent =  
  integer(0), record_mem = FALSE, inner = control_inner())
```

```
init_joint(m, d1, d2, beta = NULL, gamma = NULL, phi2 = NULL, tau2 = NULL,  
  sigma2 = NULL, theta = NULL)
```

```
fixed_joint(beta = FALSE, gamma = FALSE, phi2 = FALSE, tau2 = FALSE, sigma2 =  
  FALSE, theta = FALSE)
```

```
gibbs_unmatch(y, sigma, X, init = init_unmatch(m = length(y), d = ncol(X)),  
  control = control_unmatch(), fixed = fixed_unmatch())
```

```
control_unmatch(R = 1000, burn = 0, thin = 1, report = R + 1, save_latent =  
  integer(0), record_mem = FALSE, inner = control_inner())
```

```
init_unmatch(m, d, beta = NULL, tau2 = NULL, mu = NULL)
```

```
fixed_unmatch(beta = FALSE, tau2 = FALSE, mu = FALSE)
```

```
control_inner(tol_suff = 0.01, tol_merge = exp(-100), max_rejects = 1e+06,  
  method = c("vws-tune", "vws-basic", "imh", "arms", "amh", "mh-vws"), N = 50,  
  tune = 1e+06, amh_varprop_init = 25, amh_varprop_c = 2.4, amh_varprop_eps =  
  0.05)
```

```
traceplots(x, xlab = "")
```

3 Scripts

The following codes in `01-sim-cond` produce the conditional simulation study in Section 5.1 of the article.

- `metro.R`: Simulations with IMH.
- `vws.R`: Simulations with VWS.
- `samplers.cpp`: C++ functions for samplers consisting of one IMH step and one VWS step.

The following codes in `02-sim-post` produce the posterior simulation study in Section 5.2 of the article.

- `gen.R`: Generate folder structure for the study. Each level of the study is placed into a separate folder.
- `sim.R`: Run one level of the study.
- `analyze.R`: Analyze the results from the completed study and produce summaries.

The `03-sae` folder contains a single script `saipe.R` that produces the data analysis in Section 6 of the article.

The `data` folder contains scripts and inputs to create the example SAIPE dataset.

- `saipe.csv`: the prepared dataset.
- `cntysnap.csv`: Supplemental Nutrition Assistance Program (SNAP) data for 2022, to be used as a covariate with 2023 SAIPE data.
- `co-est2023-alldata.csv`: 2023 population estimates from the U.S. Census Bureau’s Population Estimates Program. The data may be obtained from the Census Bureau [website](#).
- `build-data.R`: Code to construct the dataset `saipe.csv`.

4 Session Information

Results in the paper were produced in an R environment with the following packages and versions.

```
library(devtools)
library(tidyverse)
library(Rcpp)
library(knitr)
library(mcmcse)
library(jsonlite)
library(xtable)
library(vws)
library(statmod)
```

```
sessionInfo()
```

R version 4.6.1 (2026-06-24)

Platform: x86_64-pc-linux-gnu

Running under: Debian GNU/Linux forky/sid

Matrix products: default

BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.1

LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.1; LAPACK version 3.12.0

locale:

```
[1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8       LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8   LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8      LC_NAME=C
```

```

[9] LC_ADDRESS=C LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: America/New_York
tzcode source: system (glibc)

attached base packages:
[1] stats graphics grDevices utils datasets methods base

other attached packages:
[1] saevws_0.2.0

loaded via a namespace (and not attached):
[1] gtable_0.3.6 jsonlite_2.0.0 dplyr_1.2.1 compiler_4.6.1
[5] brio_1.1.5 tidyselect_1.2.1 Rcpp_1.1.1-1.1 ellipse_0.5.0
[9] stringr_1.6.0 scales_1.4.0 yaml_2.3.12 fastmap_1.2.0
[13] lattice_0.22-9 coda_0.19-4.1 ggplot2_4.0.3 R6_2.6.1
[17] generics_0.1.4 knitr_1.51 tibble_3.3.1 fftwtools_0.9-11
[21] pillar_1.11.1 RColorBrewer_1.1-3 rlang_1.2.0 testthat_3.3.2
[25] stringi_1.8.7 xfun_0.59 S7_0.2.2 ote1_0.2.0
[29] mcmcse_1.5-1 cli_3.6.6 magrittr_2.0.5 digest_0.6.39
[33] grid_4.6.1 vws_0.3.0 fnt1_0.1.2 lifecycle_1.0.5
[37] vctrs_0.7.3 evaluate_1.0.5 glue_1.8.1 farver_2.1.2
[41] rmarkdown_2.31 tools_4.6.1 pkgconfig_2.0.3 htmltools_0.5.9

```

5 Adaptive Metropolis-Hastings Algorithm

An Adaptive Metropolis-Hastings (AMH) method is implemented in our studies for comparison. The particular variant (“SCAM”) is from Section 3.1 of Roberts and Rosenthal (2009) which is based on Haario et al. (2005). The objective is to draw $\mathbf{X} = (X_1, \dots, X_k)$ from target π . A chain $\mathbf{x}^{(r)}$, $r = 0, 1, 2, \dots$, produced by an MCMC method can approximate such draws. Using a Metropolis-within-Gibbs approach to draw component x_j , an “optimal” proposal distribution for Metropolis sampling would be $X_j^* \sim N(x_j^{(r)}, (2.38)^2 \text{Var}_\pi(X_j))$. We can then accept $x_j^{(r+1)} = x_j^*$ with probability

$$\alpha(x_j^*, x_j^{(r)}, \mathbf{x}_{\setminus j}) = \min \left\{ 1, \frac{\pi(x_1, \dots, x_{j-1}, x_j^*, x_{j+1}, \dots, x_k)}{\pi(x_1, \dots, x_{j-1}, x_j^{(r)}, x_{j+1}, \dots, x_k)} \right\},$$

where $\mathbf{x}_{\setminus j}$ denotes the last draws of components $\ell \neq j$ from the chain, and let $x_j^{(r+1)} = x_j^{(r)}$ with the remaining probability. The constant 2.38 arises in MCMC theory. The variance $\text{Var}_\pi(X_j)$ is computed under the target and therefore is generally not accessible except through approximation by an MCMC method. The objective of SCAM is to build up an approximation to $\text{Var}_\pi(X_j)$ within the MCMC. Roberts and Rosenthal (2009) use SCAM with the proposal $N(x_j^{(r)}, v_j^{(r)})$, where

$$v_j^{(r)} = \begin{cases} v_j^{(0)}, & r \leq r_0, \\ s^2 \{\tilde{V}_j^{(r)} + \epsilon\}, & r > r_0, \end{cases}$$

with ϵ chosen to be a small positive number, $s = 2.4$ as a rounded version of the 2.38 optimality constant, initial value $v_j^{(0)}$, and empirical variance

$$\tilde{V}_j^{(r)} = \frac{1}{r-1} \sum_{i=0}^{r-1} (x_j^{(i)} - \bar{x}_j^{(r)})^2,$$

with associated mean $\bar{x}_j^{(r)} = \frac{1}{r} \sum_{i=0}^{r-1} x_j^{(i)}$. Once the element $x_j^{(r)}$ of the chain is determined at the end of step r , the empirical mean and variance can be updated using the recursions so that it is not necessary to retain all of the draws:

$$\bar{x}_j^{(r+1)} = \begin{cases} x_j^{(0)}, & r = 0, \\ \frac{1}{r+1} \{r\bar{x}_j^{(r)} + x_j^{(r)}\}, & r > 0, \end{cases}$$

and

$$\tilde{V}_j^{(r+1)} = \begin{cases} \{x_j^{(0)} - x_j^{(1)}\}^2/2, & r = 1, \\ \frac{r-1}{r} \tilde{V}_j^{(r)} + \{\bar{x}_j^{(r)}\}^2 + \frac{1}{r} \{x_j^{(r+1)}\}^2 - \frac{r+1}{r} \{\bar{x}_j^{(r+1)}\}^2, & r > 1, \end{cases}$$

For the joint SAE application (?), we use an exponential transformation to generate positive proposal values from the normal proposal distribution for σ_i^2 . This results in the inclusion of a Jacobian term in the Metropolis ratio as follows. Let ϕ^* be a draw from proposal $N(\phi^{(r)}, v_i^{(r)})$ with $\phi^{(r)} = \log \sigma_i^{2(r)}$. The density with respect to ϕ^* (with proportionality constant omitted) can be obtained by transformation as

$$f(\exp(\phi^*) \mid -) \propto f_{\text{IG}}(\exp(\phi^*) \mid \kappa_i, \lambda_i) f_{\text{LN}}(\exp(\phi^*) \mid \mu_i, \tau^2) \exp(\phi^*),$$

where the term $e^\phi = \frac{d}{d\phi} e^\phi$ is from the Jacobian of the transformation from σ_i^2 to ϕ . The acceptance ratio for Metropolis-Hastings is then

$$\varpi_i^{(r)} = \min \left\{ 1, \frac{f_{\text{IG}}(\exp(\phi^*) \mid \kappa_i, \lambda_i) \cdot f_{\text{LN}}(\exp(\phi^*) \mid \mu_i, \tau^2) \cdot \exp(\phi^*)}{f_{\text{IG}}(\exp(\phi^{(r)}) \mid \kappa_i, \lambda_i) \cdot f_{\text{LN}}(\exp(\phi^{(r)}) \mid \mu_i, \tau^2) \cdot \exp(\phi^{(r)})} \right\}. \quad (1)$$

For the unmatched SAE application (You and Rao 2002), we draw proposal variate $\mu_i^* \sim N(\mu^{(r)}, v_i^{(r)})$ and accept with probability

$$\varpi_i^{(r)} = \min \left\{ 1, \frac{f_{\text{LN}}(\mu_i^* \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\text{N}}(\mu_i^* \mid y_i, \sigma_i^2)}{f_{\text{LN}}(\mu^{(r)} \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\text{N}}(\mu^{(r)} \mid y_i, \sigma_i^2)} \right\}. \quad (2)$$

Note that the Gaussian proposal cancels from both (1) and (2) because of symmetry.

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